UNIVERSITY OF BIRMINGHAM

Forensic

Computational Authorship Analysis

Dana Roemling (they/them)
26.06.25

Today

- Session 1.1 – 14.00-15.30
  - Intro to Forensic Linguistics
  - Intro to Authorship Analysis
- Session 1.2 – 15.45-17.00
  - Practical Application

⚠
Content warning

Discussions of death, murder, assault, ransom/kidnapping

Mostly no details, but manner of death is discussed in one case

Forensic Linguistics

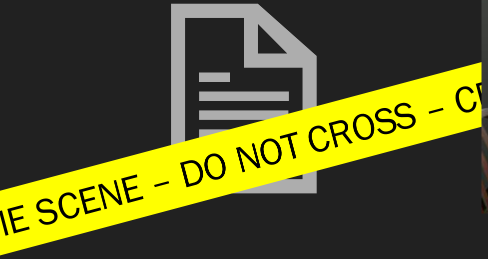
Forensic Linguistics

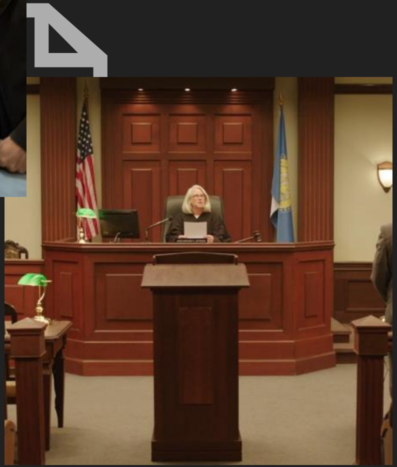
- A branch of applied linguistics
- The intersection of language and legal or investigative contexts
- Specialist field closely linked to practical work of law enforcement, crime investigators, prosecutors, judges, ...
- Established mainly in English-speaking countries, but a growing area of research in many other jurisdictions as well

Forensic Linguistics

- The application of linguistic methods, approaches and knowledge to investigative, criminal or legal contexts
- Language as evidence or language of the legal process, language and law

Forensic Linguistics





Investigative

Legal

Forensic Linguistics

- Language as evidence & language in the investigative process
    - Application of linguistics to criminal cases and investigations, e.g., authorship analysis or threatening communication
    - for example: ransom note, text messages in a murder investigation, but everything could be a forensic text

Forensic Linguistics

- Language of the law & language in the legal process
  - Laws, statutes, guidelines
  - Writing, interpretation, application of these texts
  - Translation, interpreting
  - Normative language use in court, legal terminology
  - Legalese & implications for legally untrained people
  - for example: statutory interpretation, the influence of power & hierarchy in court

Forensic Linguistics

- ≠ Forensic Phonetics
- Methodologies in FP are quite different
- Analysis of Spoken Language
    - Identifying voice and speaker characteristics
    - Speech recognition and authentication
    - Determining who spoke
    - Voice comparisons
    - "Voice Line-Ups" for speaker identification
    - "Voiceprint" analysis for unique vocal patterns, profiling

Forensic Linguistics

- Syntax/Morphology – Sentence structure and word formation
- Semantics/Pragmatics – Meaning and language use in context
- Sociolinguistics – Language in social and cultural contexts
- Discourse Analysis – Examining language in communication
- Language Change – Evolution of language over time
- Dialectology – Study of regional and social language variation
- Psycho-/Neurolinguistics – Language processing in the brain
- Computational Linguistics – Language and technology
- Corpus Linguistics – Data-driven linguistic analysis
- Genre, Register, Grammar & Orthography – Conventions of structure and spelling

AUTHORSHIP ANALYSIS

Authorship Analysis

- Authorship generally means being the author of a (linguistic) work, however:
- Linguistic authorship analysis aims to identify the writer of a disputed text by examining language patterns
- It focuses on textual elements such as vocabulary, grammar, style, and spelling rather than the subject matter
- The analysis excludes non-linguistic evidence like handwriting or external clues such as fingerprints or IP addresses
- The goal is to determine the author of a document based on writing style

Forensic Authorship Analysis

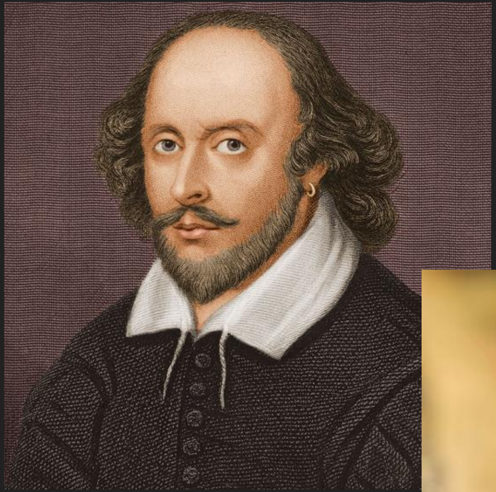
- The scientific study of written language to determine the identity or characteristics of an author in legal or investigative contexts
- Aims to verify/determine authorship by analysing linguistic and stylistic patterns
- Examines features such as word choice, grammar or sentence structure
- Based on the principle that each writer has a unique way of using language, shaped by habits, education, and personal style

Authorship Analysis

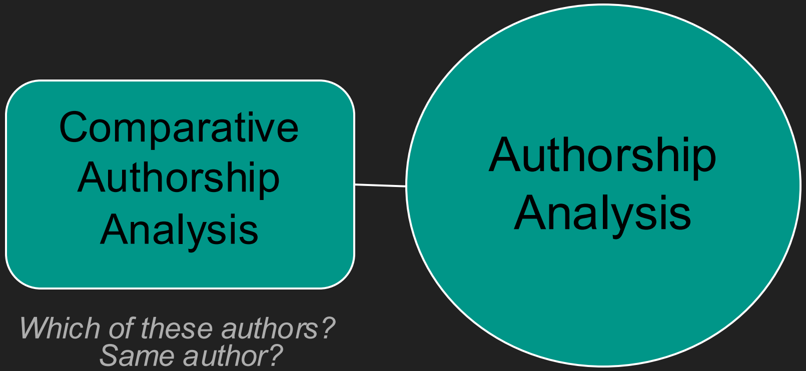
- Who wrote it?
- When was it written?
- Was it translated? Was it copyedited?
- Was it plagisarised?
- Was there more than one author?

- **Authorship analysis** has a long history, with quantitative methods dating back to the 19th century

Authorship Analysis

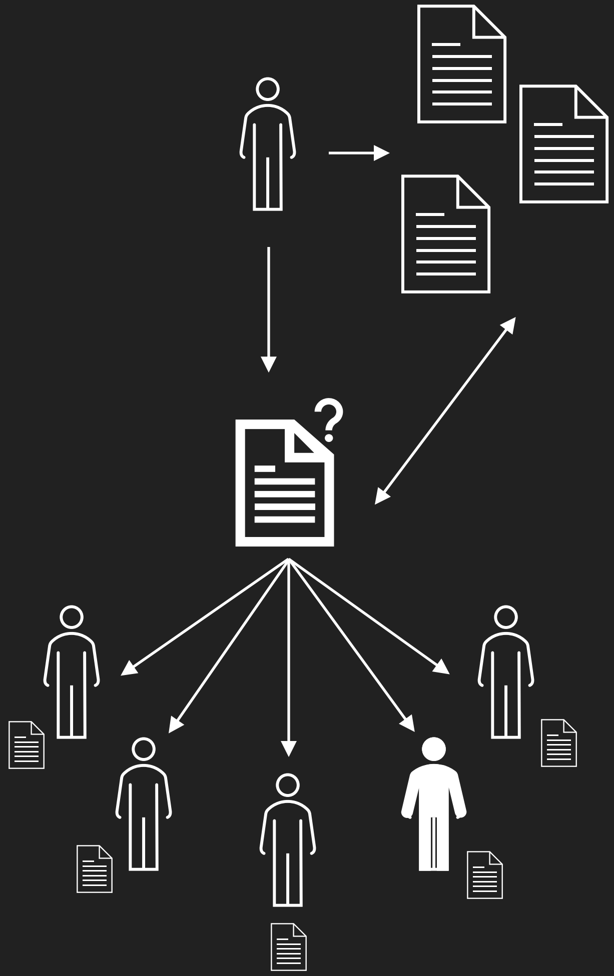


Authorship Analysis

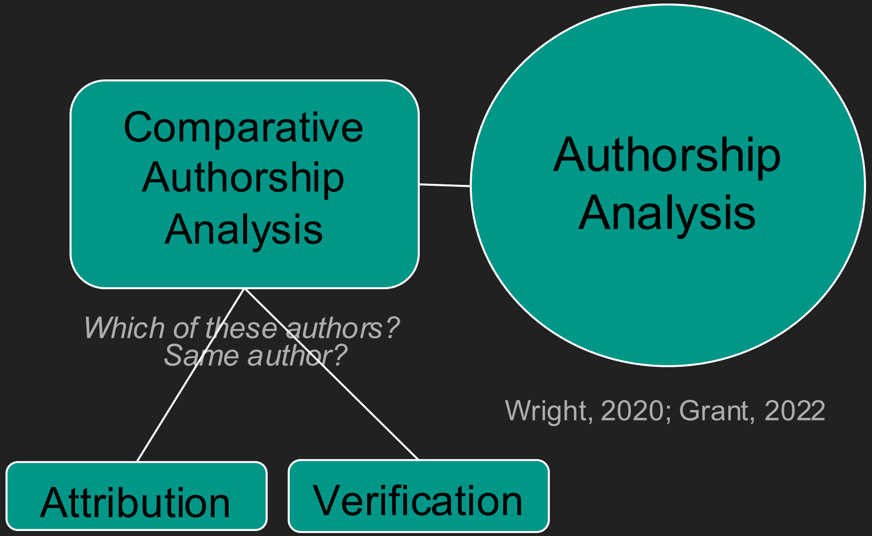


Which of these authors?
Same author?

Wright, 2020; Grant, 2022

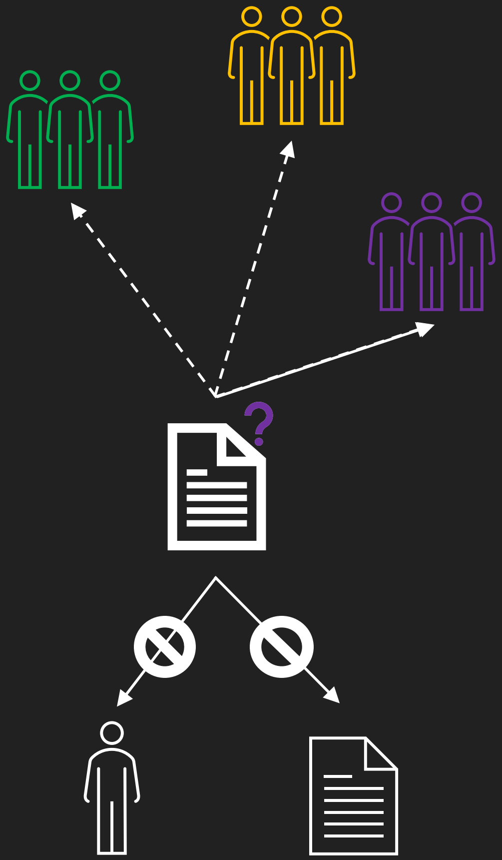


Authorship Analysis

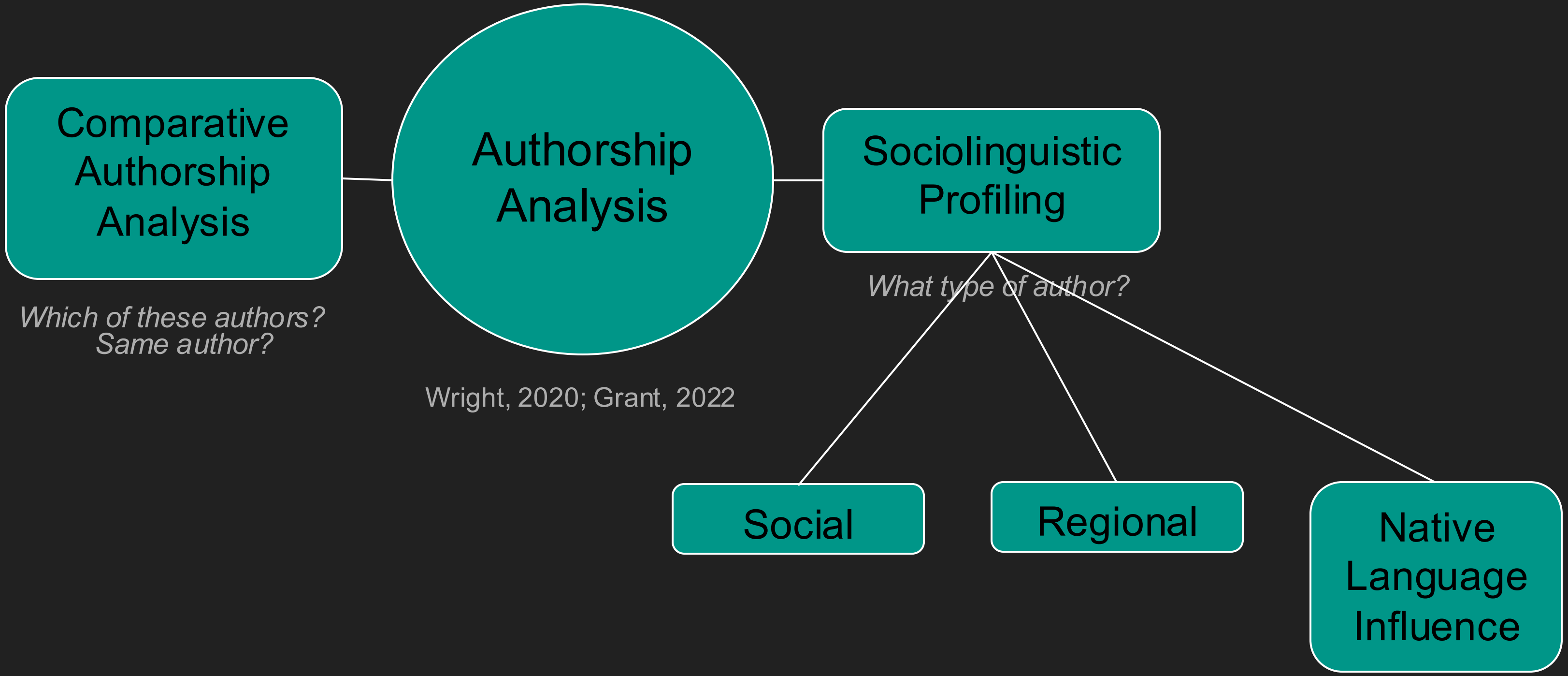


Authorship Analysis

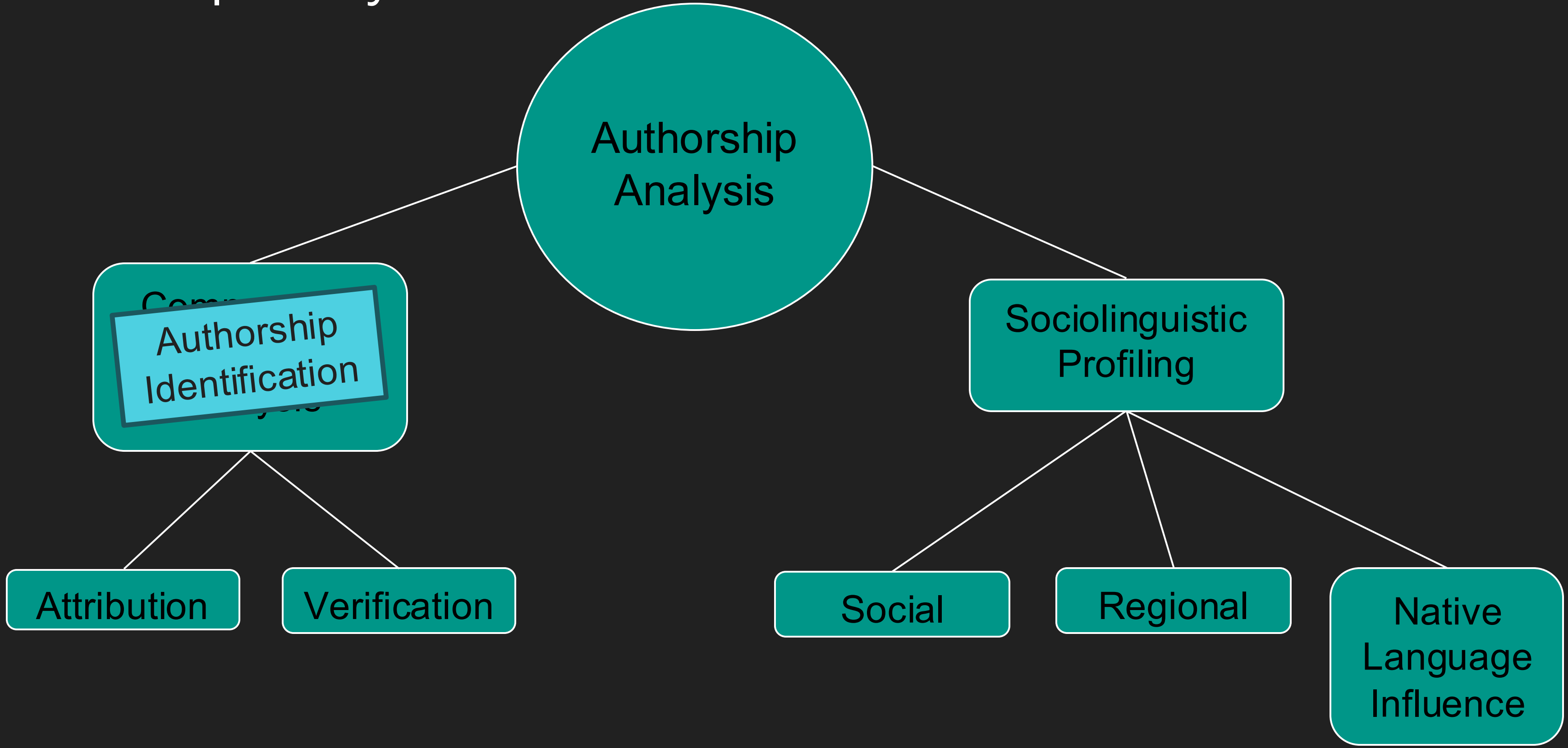
Wright, 2020; Grant, 2022



Authorship Analysis



Authorship Analysis



Wright, 2020; Grant, 2022

Forensic Authorship Analysis

Comparative Authorship Analysis / Authorship identification has two main types of tasks: **authorship attribution** and **authorship verification**

- **Authorship Attribution (Closed Set):** Determines which of several candidate authors has the most similar writing style to the questioned document
- **Authorship Verification (Open Set):** Assesses whether a given author wrote the questioned document
- Verification is more challenging than attribution and is considered a broader problem

Forensic Authorship Analysis

While identification and profiling are the most common authorship analysis tasks, there are several others

- **Authorship Clustering:** Grouping texts by author without prior knowledge of the number or identity of authors
- **Authorship Obfuscation:** Altering writing style to conceal authorship
- **Chronological Analysis:** Sorting an author's texts by date or tracking stylistic changes over time
- The field of authorship analysis continues to evolve, including emerging tasks like AI-generated text detection

Idiolect

„[...] every native speaker has their own distinct and individual version of the language they speak and write, their own idiolect [...]“

„ [...] assumption that idiolect will manifest itself through distinctive and idiosyncratic choices in texts“

(Coulthard, 2004)

Idiolect

„[…] in principle any speaker/writer can use any word at any time, [but] speakers in fact tend to make typical and individuating co-selections of preferred words"

(Coulthard, 2004)

→ There is enough linguistic variation between authors to allow for linguistic features to link to one author rather than the other

Idiolectal style

- Marko et al. (2022) opt for calling this variation ‘individual style’
- Grant and MacLeod (2018) propose that the variation used in authorship analysis should be considered identity performance, rather than idiolect
- Grieve (2023) argues that idiosyncratic features are actually features of author register variation, rather than of individual authors
- Nini (2023) summarises that for computational analysis we usually assume that individuals differ in how frequently they use certain words in certain functions (what he calls “functors”) and that we use language in chunks (and not just as single words); these units are entrenched/conventionalised and then form the grammar(s) we use

Authorship methodologies

- In comparativ ocuments are examined by comparing xts
- Multiple lingui y consistencies and inconsistencie
- This comparis ss different tasks

Common features include:

- Word use (esp. function words)
- Character sequence use (n-grams)
- Word sequences use (n-grams)
- Word and sentence length
- Vocabulary richness
- Part-of-speech use
- Punctuation mark use

Authorship methodologies

- **Qualitative Authorship Analysis:** Involves close reading to identify categorical patterns, such as whether an author uses a specific feature or not. Best for small datasets
- **Quantitative Authorship Analysis:** Uses statistical methods to measure and compare feature frequencies, often with pre-selected feature sets. Best for large datasets

Authorship methodologies

A distinction is often made between **stylistic (qualitative)** and **stylometric (quantitative)** studies

- **Stylometry** is the quantitative study of authorial style (Holmes 1994, Stamatatos 2009). Most stylometric research focuses on **comparative authorship analysis**, particularly attribution, but it also covers profiling, clustering, obfuscation, and chronological analysis
- Stylometry dominates authorship analysis research, though stylistic analysis remains important, especially in literature and forensic linguistics. Modern **LLM-based approaches** can also be classified as stylometric

Authorship methodologies

Computational

- Quantifying linguistic style
- Sentence length, word length, paragraph length, text length, word frequencies
- Repetition and lexical richness
- Type-Token-Relation
- N-grams, bag-of-word, neural approaches
- Statistics

(Grant & Baker 2001)

Manual

- Analysis of stylistic variation
- Collocations, phrases, lexical features
- Hapaxes (*Hapax legomenon*)
- Punctuation
- Variance & style

(McMenamin 2001)

Grant & Grieve (2022): The Starbuck Case

Authorship methodologies

Authorship analysis research is spread across multiple domains, each with distinct techniques. Researchers generally come from three main areas:

- **Literature** (including Digital Humanities and some historians)
- **Linguistics** (Forensic Linguistics, Discourse Analysis, Stylistics, Corpus Linguistics)
- **Computer Science** (NLP, Machine Learning, AI, Computational Linguistics)
- The field is also attracting **data scientists, journalists, and cybersecurity analysts**

Authorship methodologies

While some forensic linguistics researchers use quantitative methods, the field remains a primary hub for qualitative authorship research.

This is largely because forensic linguistic texts, both questioned and comparison documents, are typically short and high-stakes, requiring careful analysis.

Authorship methodologies

- **Authorship attribution** is now considered reasonably accurate for cases that meet specific criteria:
  - The questioned document is **relatively long** (at least 500–1,000 words)
  - There is a **small set of candidate authors** (typically 2–5)
  - Each author has a **substantial amount of known writings** (at least 5,000–10,000 words)
  - **Genre and register** are controlled across all texts

Authorship methodologies

In cases where data is not ideal, there's still a lot of work to do:

- Short Questioned documents
- Limited writing samples
- Questioned and Known Writings in different register/genres
- Authorship Verification
- Authorship Profiling
- Methods for feature selection and comparison

Who is the author?

Oh thanx! - were meeting at 4 and then gotta go 2 jamies after. The party sounds brilliant! Have a gd day! xx

Author A

Okay, cool! - but i think it actually starts at 5, so we wont make the first half, were prob bk about half 6?

Yeah he said that but hasnt given me anything else to work with so i hope nothing changed x

Right thanks! i'll make sure to see you then- still some time! be safe 2 xx

Author B

Okay thats good to know. Do they start setting up on the 13th?

Hey hey, how did it go? I ended up going to the game with marc last night which was really enjoyable!

I can't tell you exactly, marc's sorted all that out and mentioned nothing about that. When did you say you're leaving? Xxx

Case

- Amanda Birks (AB)
- Christopher Birks (CB)

- Following a house fire, the police cannot rule out foul play
- The body of AB does not show injuries typical of fire (ante mortem), neither does it show a cause of death typical of fire
- Could it be that AB was already dead, potentially murdered?

Case

- Involved a murder investigation where authorship of SMS messages was central to the case
- Disputed authorship of a series of text messages sent from the victim's phone after her presumed time of death
- Did the victim send the messages herself - or did someone else (the suspect) send the messages to create an alibi?
- Comparing linguistic features of known vs. questioned messages.

Grant, T. (2013). TXT 4N6: Method, Consistency, and Distinctiveness In the Analysis of SMS Text Messages. *JOURNAL OF LAW AND POLICY*, *21*(2), 467–494.

Assumptions (Grant, 2013)

- Comparative authorship analysis relies on two key theoretical assumptions:
  - Assumption 1: An individual author’s writing style is consistent across their relevant texts.
  - Assumption 2: This consistent style is distinctive enough to differentiate that author from others.

Methodology

- Text messages are short and running automated analyses like we could when comparing books was not an option here
- Instead, Grant relied on consistency of features to create stylistic profiles of the known authors, so before there was doubt about the authorship of text messages

- Searched for recurring stylistic features across texts
- Assessed intra-author consistency (does the same person write in a consistent style?)

Features

- Orthography (e.g., capitalisation, spelling choices)
- Punctuation patterns
- Lexical choices (e.g., common phrases)
- Grammatical constructions

- Needed to be frequent enough that consistency could be assessed

Style consistency

- Features in 10 or fewer messages disregarded
- Features needed to occur double the amount of time for one author than the other (so 66% of all occurrences needed to be with one author)

  → 18 features remain for analysis

Distinct markers of authorial style for both AB and CB

'ad' for 'had' (used only by AB).

‘don’t’ for ‘don’t' (used only by AB).

‘t’ for 'the' (rare use by CB).

‘bak' for 'back' (rare use by CB).

'av' for 'have' (rare use by CB).

‘wud' for 'would' (rare use by CB).

‘w' for 'with' (some use by CB).

'wil' for 'will' (some use by CB).

‘y’ for ‘yes’ (some use by CB).

‘wen’ for ‘when’ (some use by CB)

‘2’ for ‘to’ with no trailing space (used only by CB).

‘4’ for ‘for’ with no trailing space (used only by CB).

'wiv for 'with' (used only by CB).

'jst' for 'just' (used only by CB).

‘dnt’ for don’t (used only by CB).

commas (rare use by AB).

'4get' for 'forget' (rare use by AB).

'thanx' for 'thanks' (rare use by AB).

AB is seen alive at 11AM

10:04	Friend 1	Got go fetch milly. Val cant cope w her x	Contains “w” for “with,” identified as a feature of AB’s style. CB tends to use “wiv” but does use “w” on occasions.

The police assume CB takes over the phone at 12.07PM

12:07	Friend 1	Txt me, talkin with chris. X x	Contains a comma, which is rare in AB’s texts. Slight inconsistency with AB, consistent with CB.

Analysts see distinct style change at 12.39PM

12:39	Friend 1	Am talkin wiv chris, am confused. Ur 2 young 4me. X x	Contains the use of commas, the use of “wiv” rather than “w” or “with” and the use of “4” without a trailing space. ***First text to be judged inconsistent with AB and consistent with CB.***

Following messages consistent with CB

14:03	Friend 1	Dnt kno, need think, am goin relax in bath then go sleep, really tired. X	Contains commas and "dnt." ***Judged inconsistent with AB; consistent with CB.***
17:32	Friend 2	Sorry just want time 2think. Been talkin 2chris so tryin get my head 2getha! R u out with wayne 2nite? X	Contains "2" without a following space. ***Judged inconsistent with AB; consistent with CB.***
17:37	Friend 2	Thanx tim, i just want chris talk 2me right, he needs learn not take his stresses out on me, then we can b happy. X	Contains commas and the use of "2" without a following space. ***Judged inconsistent with AB; consistent with CB.***

Results

- Clear stylistic differences identified between the victim’s known messages and the disputed texts
- Victim’s texting style showed consistent patterns:
  - Specific abbreviations and spelling habits
  - Consistent punctuation use
  - Characteristic grammar and phrase choices
- Disputed messages lacked these consistent features, showing different linguistic habits
- Evidence suggested the disputed texts were not authored by the victim

Wright, 2020; Grant, 2022

“Do you ever want to see your precious little girl again? Put $10,000 cash in a diaper bag. Put it in the green trash kan on the devil strip at corner of 18th and Carlson. Don’t bring anybody along. No kops!! Come alone! I’ll be watching you all the time. Anyone with you, deal is off and dautter is dead!!!”

(Shuy, 2001, p.1)

"Do you ever want to see your precious little girl again? Put $10,000 cash in a diaper bag. Put it in the green trash kan on the devil strip at corner of 18th and Carlson. Don't bring anybody along. No kops!! Come alone! I'll be watching you all the time. Anyone with you, deal is off and dautter is dead!!!"

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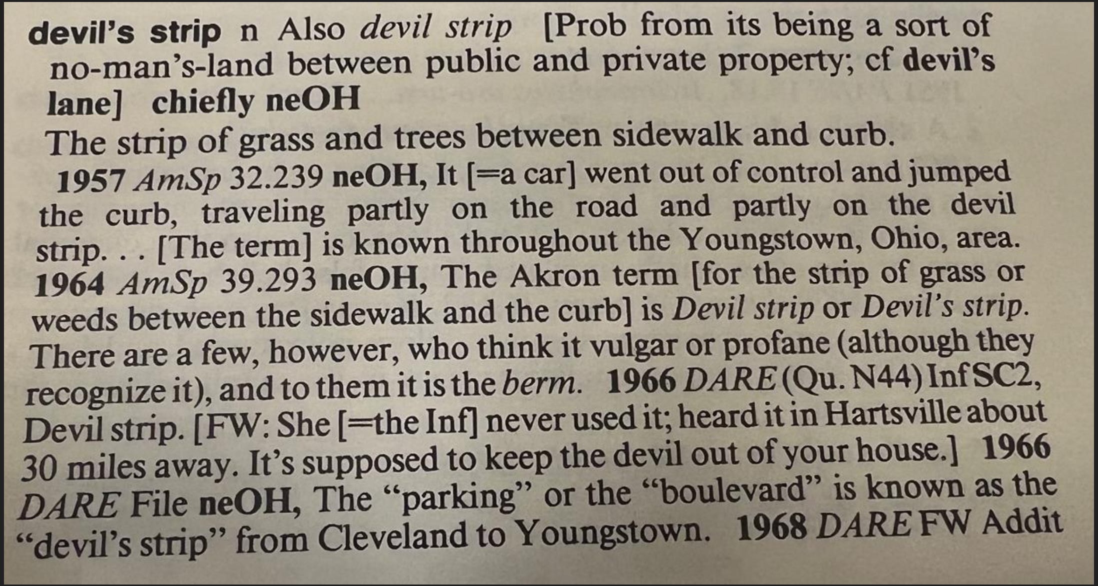
Devil Strip



devil's strip n Also *devil strip* [Prob from its being a sort of no-man's-land between public and private property; cf **devil's lane**] **chiefly neOH**

The strip of grass and trees between sidewalk and curb.

1957 *AmSp* 32.239 **neOH,** It [=a car] went out of control and jumped the curb, traveling partly on the road and partly on the devil strip. . . [The term] is known throughout the Youngstown, Ohio, area. **1964** *AmSp* 39.293 **neOH,** The Akron term [for the strip of grass or weeds between the sidewalk and the curb] is *Devil strip* or *Devil's strip.* There are a few, however, who think it vulgar or profane (although they recognize it), and to them it is the *berm.* **1966** *DARE* (Qu. N44) Inf **SC2,** Devil strip. [FW: She [=the Inf] never used it; heard it in Hartsville about 30 miles away. It's supposed to keep the devil out of your house.] **1966** *DARE* File **neOH,** The "parking" or the "boulevard" is known as the "devil's strip" from Cleveland to Youngstown. **1968** *DARE* FW Addit

Dictionary of American Regional English, "devil's strip", 1985

Authorship Profiling

- “[D]etermining the characteristics of an anonymous author, such as their demographic details, from the way they use language”
- The idea is that this can narrow down the pool of suspects
- This is usually done through either:
  - The analysis of salient linguistic features *or*
  - The analysis of writing style

(Nini, 2018)

The Analysis of Writing Style

- “[O]ften involves the study of the frequency with which certain features are used, like the study of register variation and takes as the unit of analysis the text itself”
- “A style is a variety of language associated with a particular author or social group [...] which is constituted by linguistic features that are pervasive and frequent”
- Problem: “Computational authorship profiling is not necessarily interested in understanding the inner (linguistic) mechanisms of the machine, as long as the accuracy rates are outperforming previous models.”

(Nini, 2018)

The Analysis of Salient Linguistic Features

- “[T]he application of sociolinguistic knowledge on a case by case basis to extract *ad hoc* linguistic features that are markers of a certain demographic background”
- “[I]nvolves the linguist’s experience in discovering dialectal or sociolinguistic features that can reveal clues about the background of the author”
- Problem: ”[R]elies almost entirely on the knowledge and intuition of the forensic linguist”

  (Nini, 2018)

(Regional) Dialect

- Non-linguists often use *dialect* to refer to a regional variety of a language, including a certain judgement alongside the word *dialect*
- In linguistics it refers more generally to a variety that is different from another variety of language
- The difference between the varieties might be regional in nature, but might also be, for example, socio-economic or level of formality

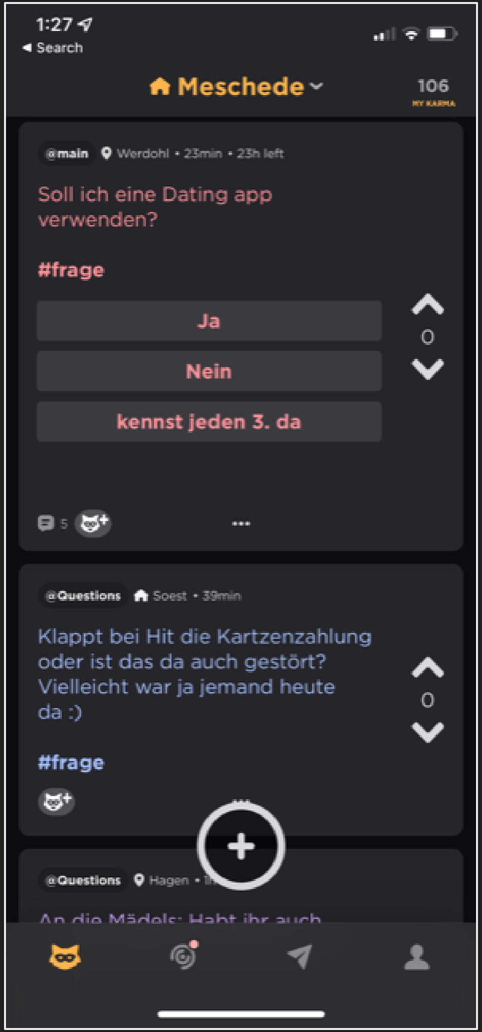
(Chambers & Trudgill, 1998)

Regional Authorship Profiling

- Based on the expert's knowledge of regional variation
- Based on previous work in dialectology / sociolinguistics
  - Works with elicited, often not digitised data (surveys, interviews)
  - Data takes considerable time to be gathered and analysed and is often outdated given how rapidly language and regional variation changes
- But we can use social media data and geostatistical methods

The Jodel Corpus

- Jodel: Social Media app
- Interaction in a 10-15km radius around own location
- Collected 2017 by Hovy & Purschke (2018) to represent German-speaking area (= Austria, Germany, Switzerland)
- No accounts / profiles, so “anonymous” interaction
- The corpus has 239,151,815 tokens at ~8000 unique locations in the GSA



The Jodel Corpus: Sample

Message	Creation Timestamp	Location	Post ID
Semesterferien: grillen schlafen grillen bar schlafen repeat.. English: *semester break: bbq sleep bbq pub sleep repeat..*	2017-04-04T 22:29:41.814Z	Berlin	58e41e5512e80 a3f0cb6f66b
Ich bin grade leicht verwirrt Werdet ihr Mädels so emotional, kurz bevor ihr eure Tage habt, oder mittendrin? English: *I am slightly confused now Are you girls being emotional just before you're on your periods or in between?*	2017-04-04T 22:04:48.109Z	Hamburg	58e41880a149d 37f12cca9d1

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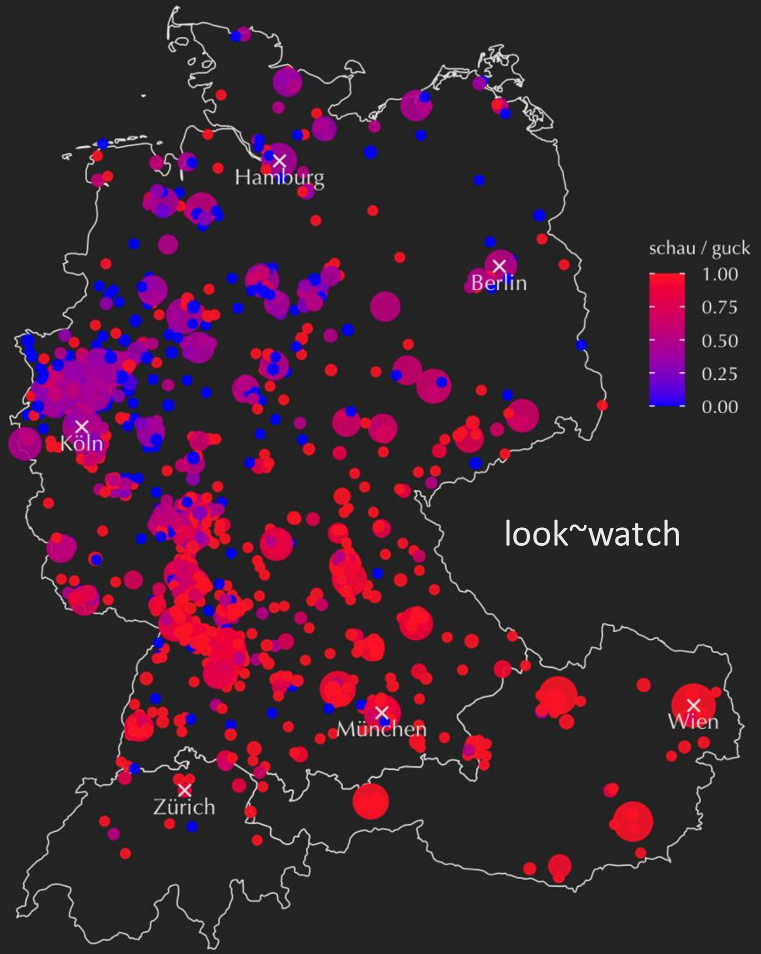
Hamburg
Berlin
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München
Wien
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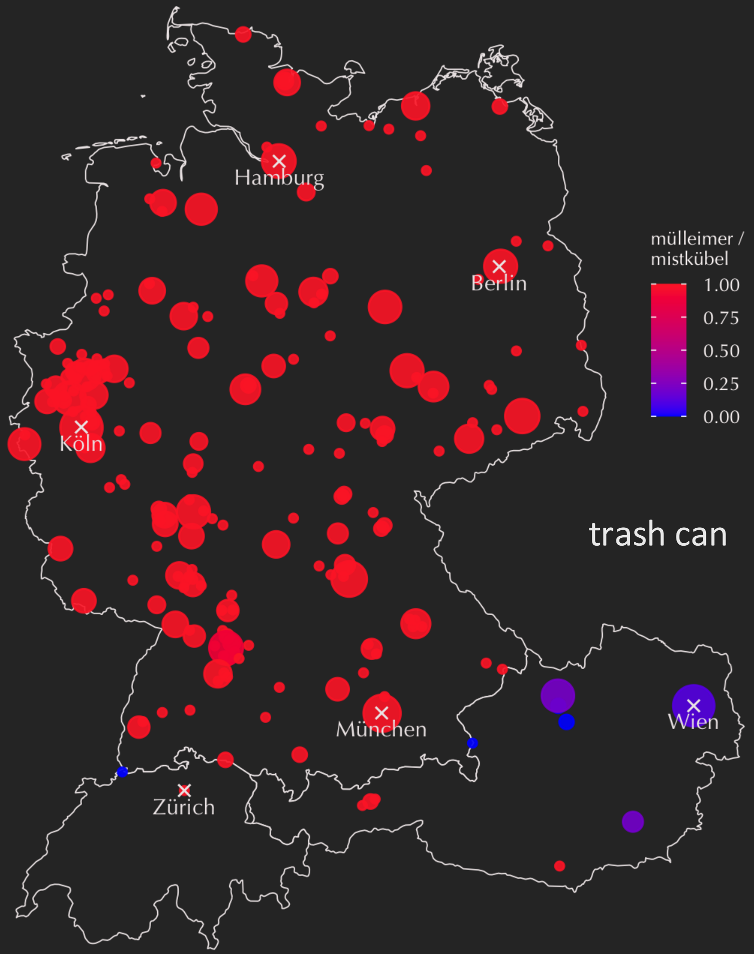
0.00
look~watch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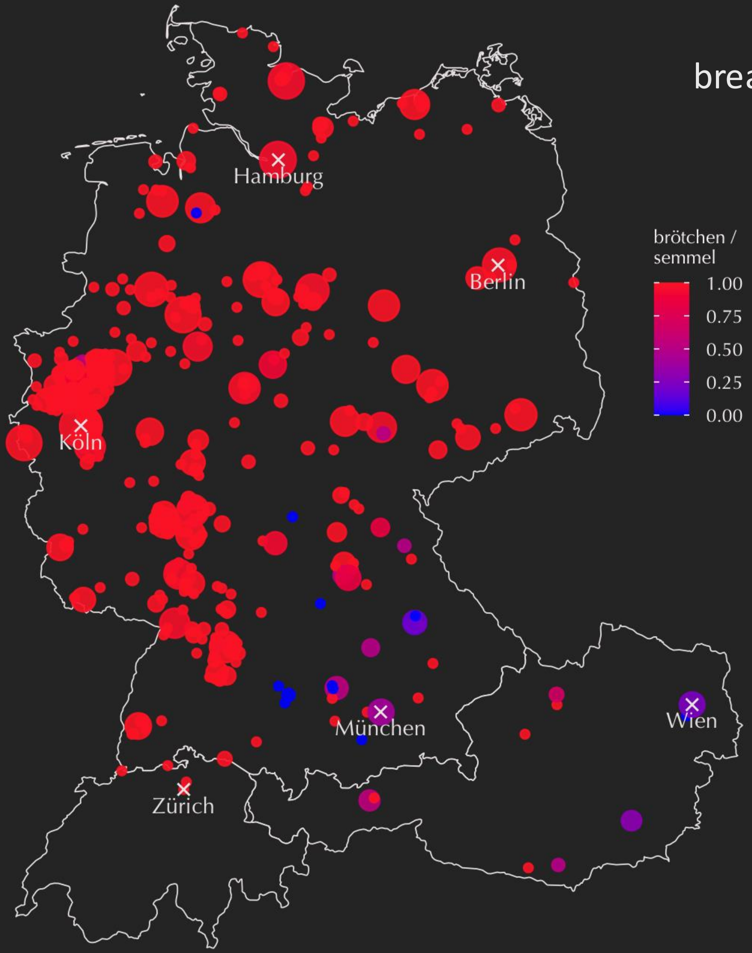
bread roll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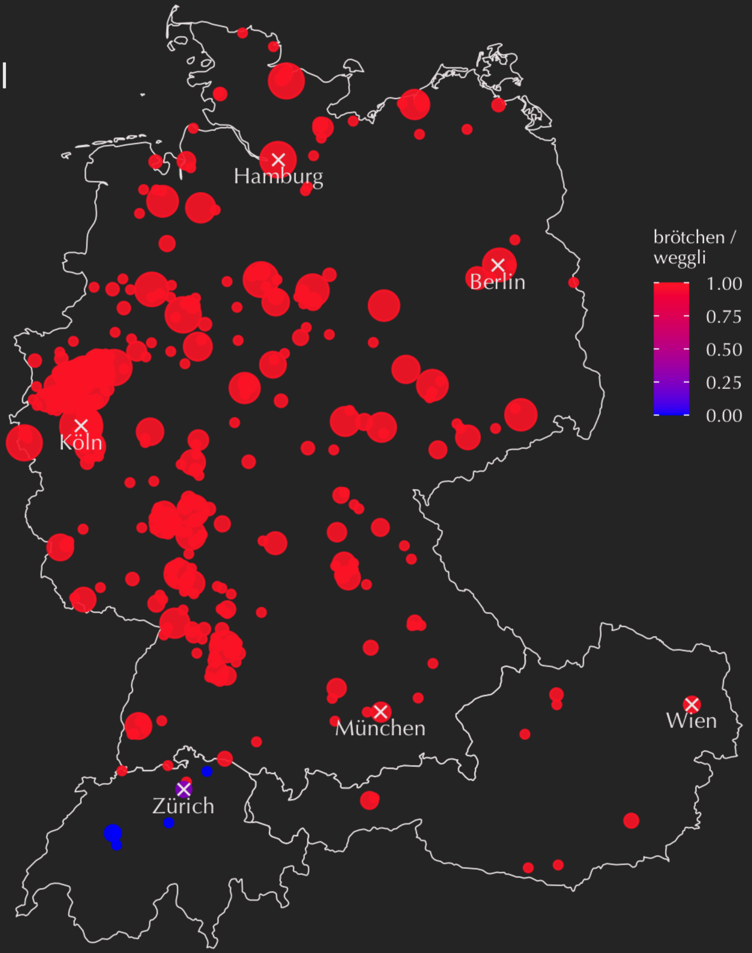
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0.00

How to test for regional distribution without prior knowledge?

- Using the 2000 most frequent tokens in the Jodel corpus
- At locations with at least 10,000 total tokens
- Calculating Getis-Ord z-scores to understand regional variance

- Using a Principal Component Analysis to reduce dimensions and understand which tokens are more relevant for the dialect / cultural regions

Louf, T., Gonçalves, B., Ramasco, J. J., Sánchez, D., & Grieve, J. (2023). American cultural regions mapped through the lexical analysis of social media. *Humanities and Social Sciences Communications*, *10*(1), 133. https://doi.org/10.1057/s41599-023-01611-3

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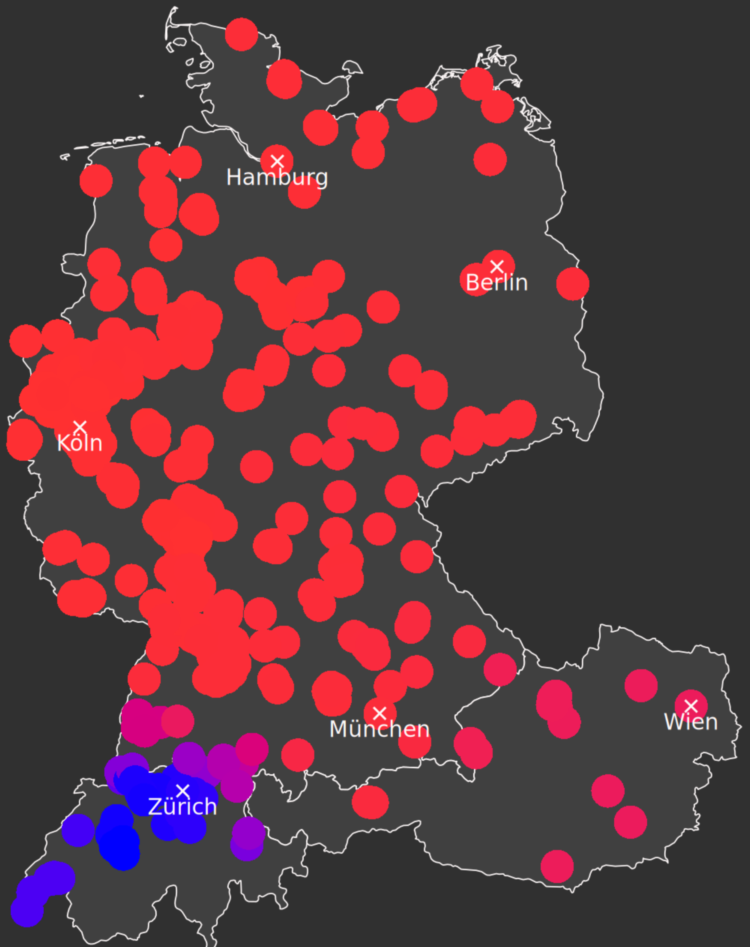
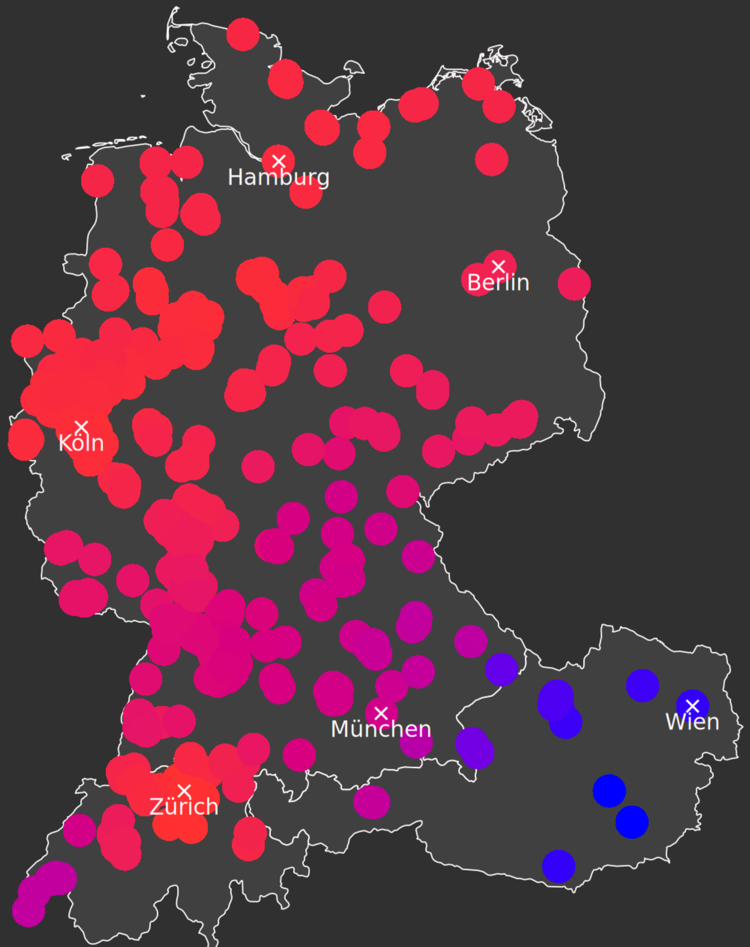
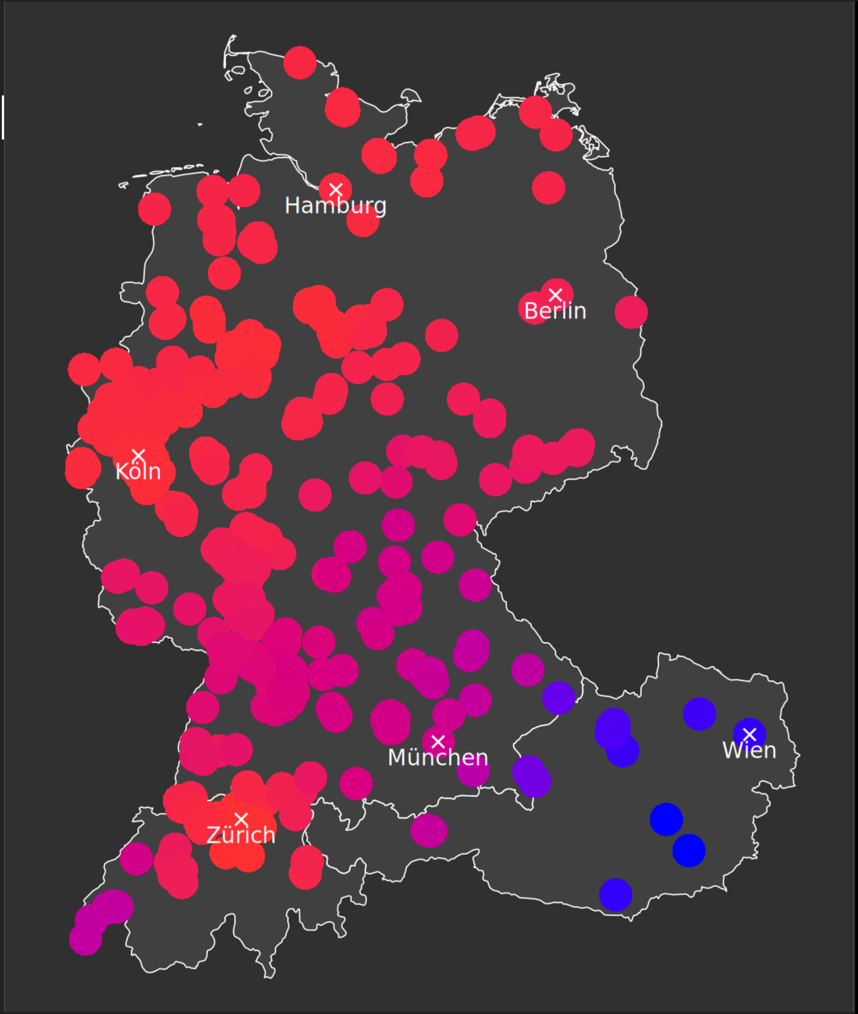
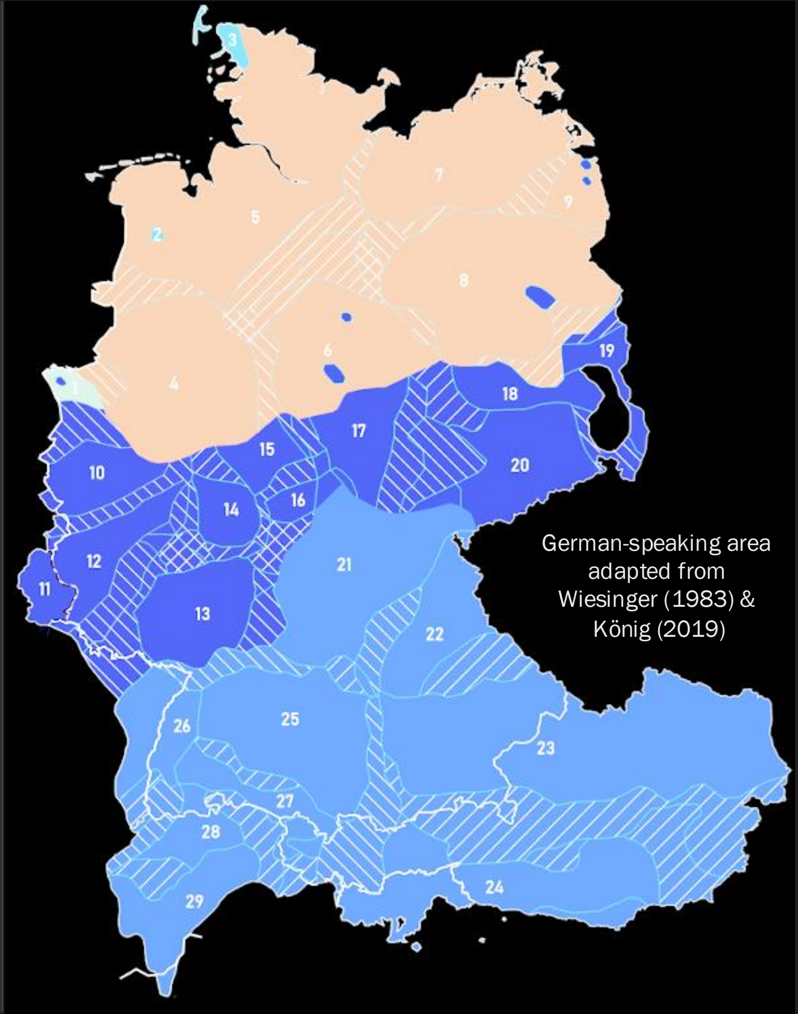


Dimension 1



Dimension 3





German-speaking area adapted from Wiesinger (1983) & König (2019)

Ordinary Kriging

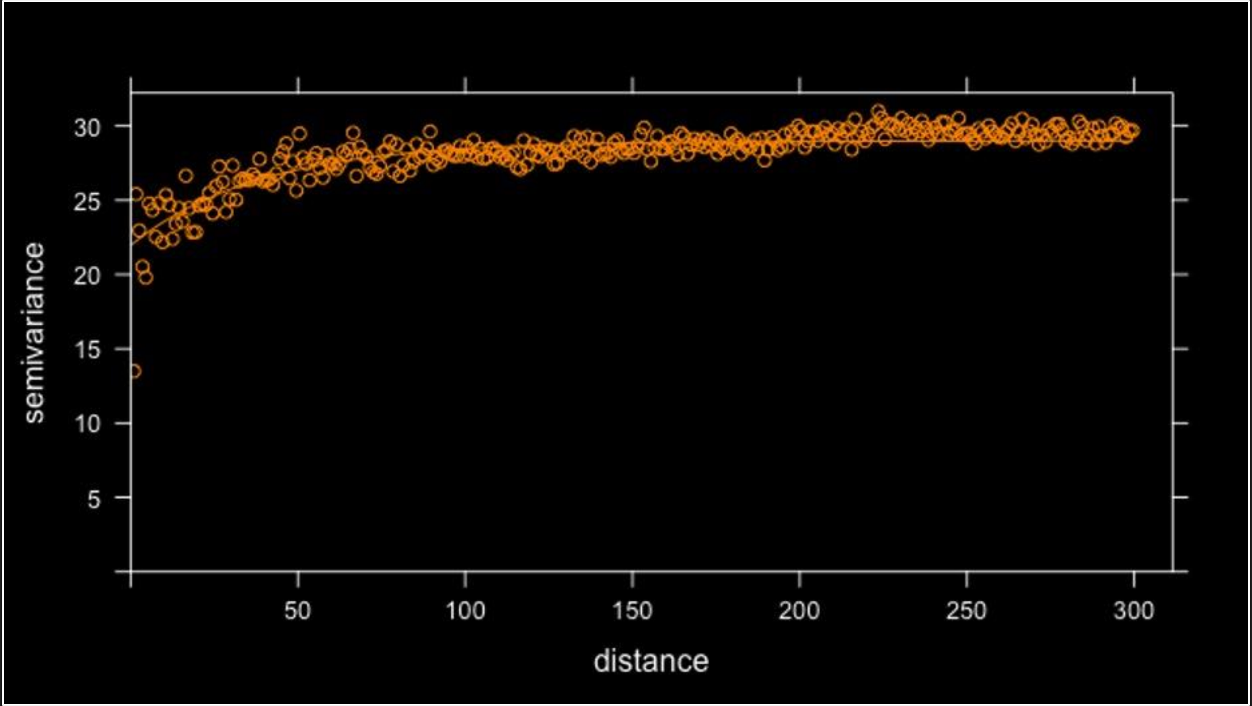
Accounting for unobserved locations

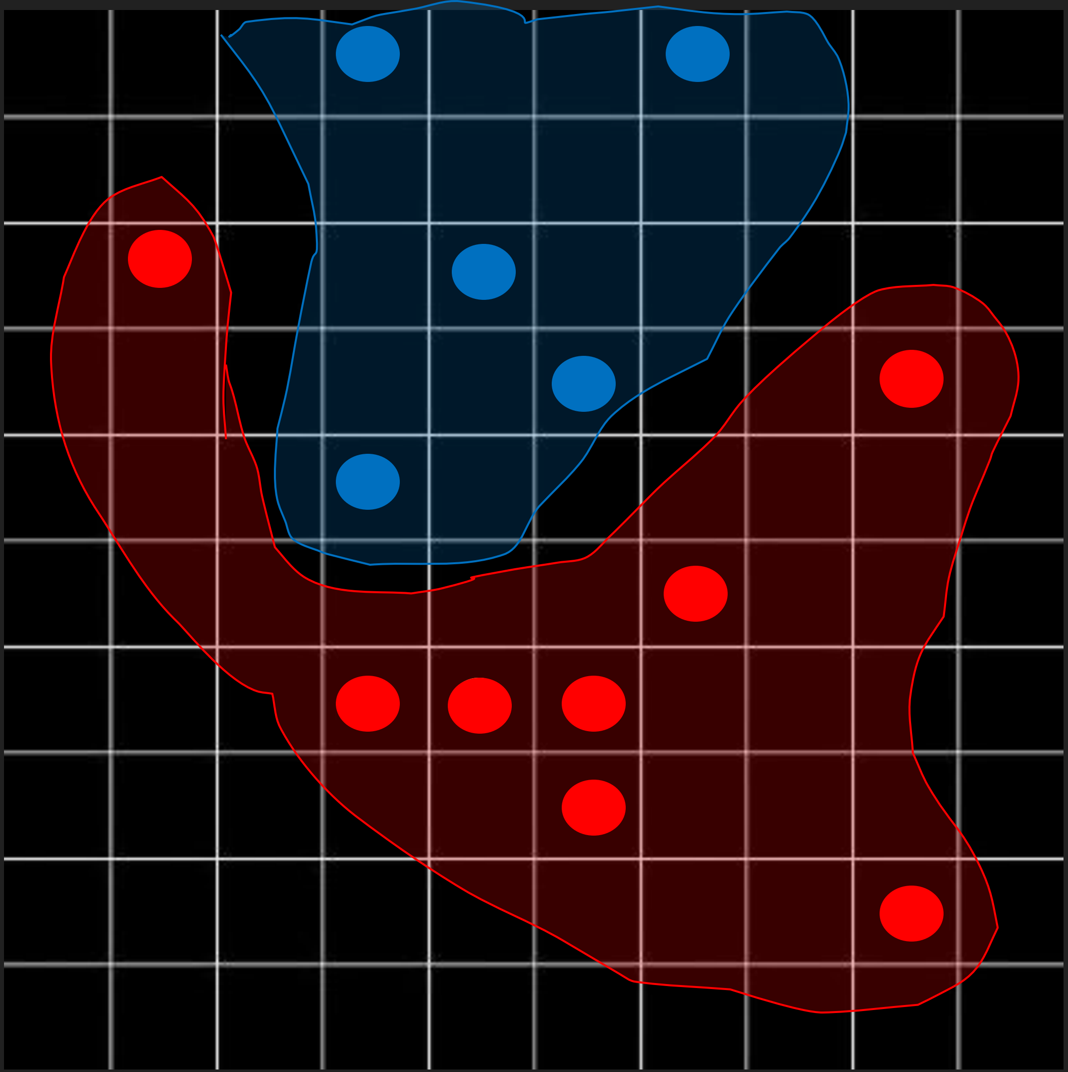
Similarity

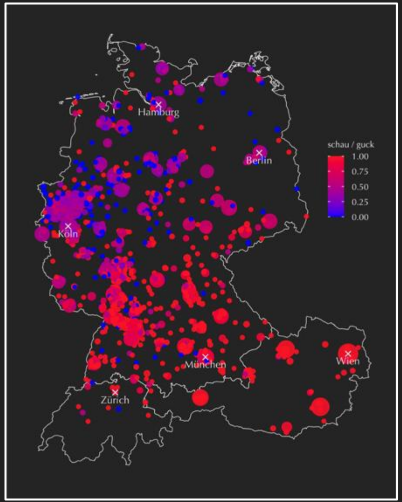
Heeringa & Nerbonne 2001
(see Nerbonne et al. 2005)

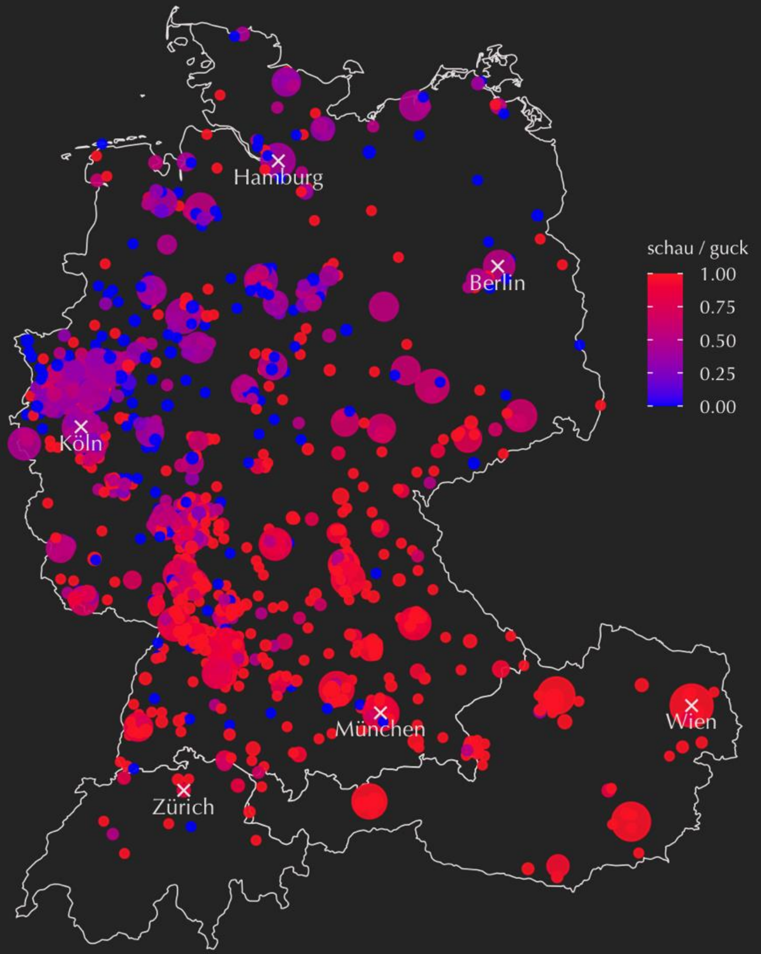
FIGURE 4. Variants of *zijn* 'to be' in IPA.

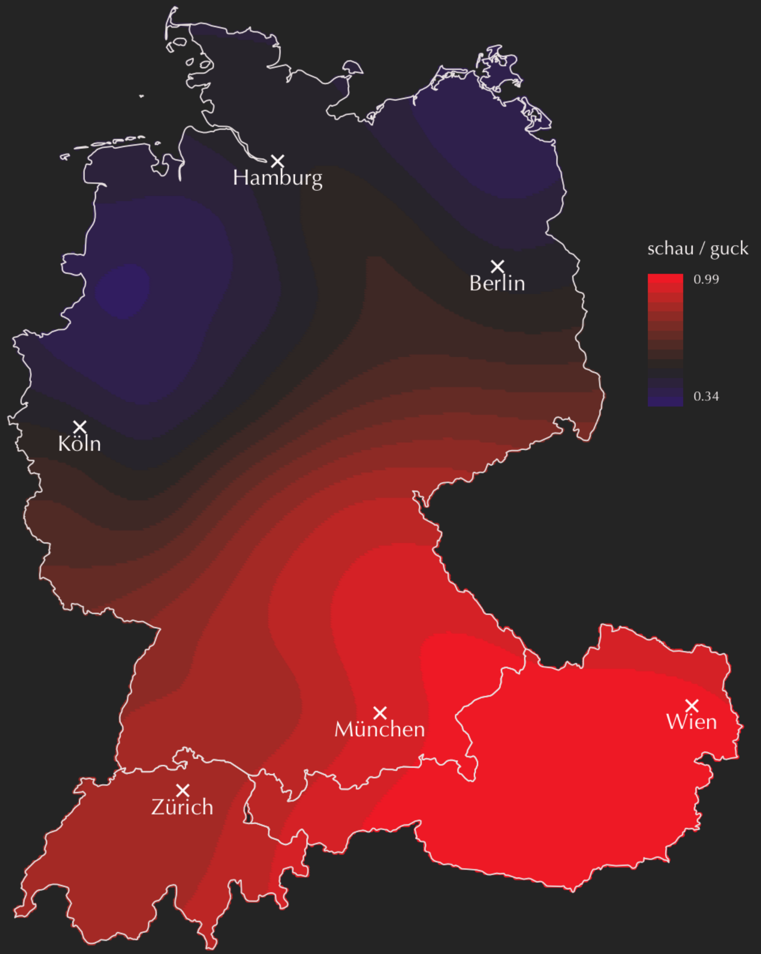
Similarity & Distance






Hamburg
Berlin
Köln
München
Zürich
Wien
schau / guck
1.00
0.75
0.50
0.25
0.00


Hamburg
Berlin
Köln
München
Wien
Zürich
schau / guck
1.00
0.75
0.50
0.25
0.00

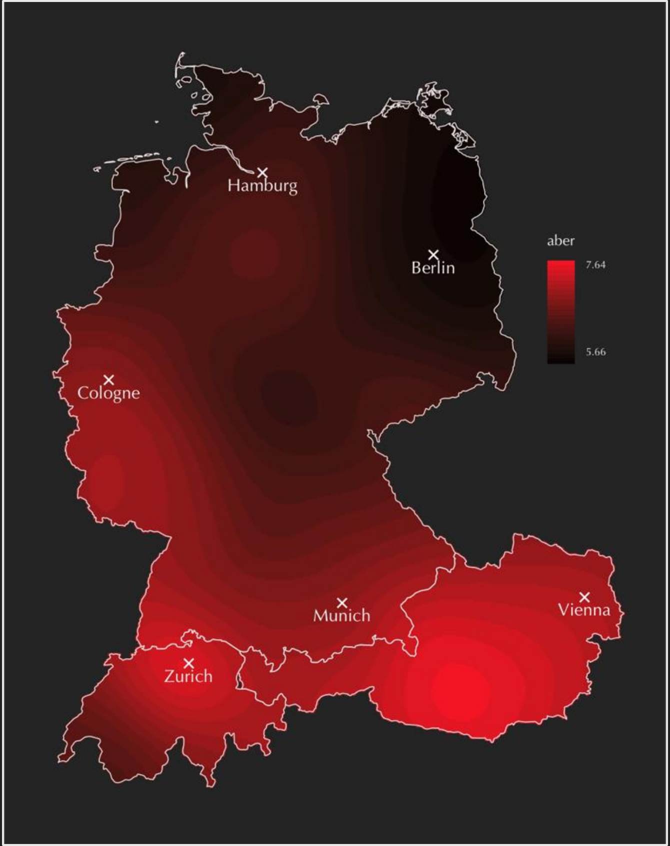

Hamburg
Berlin
Köln
München
Wien
Zürich
schau / guck
0.99
0.34

Using kriged maps for location prediction

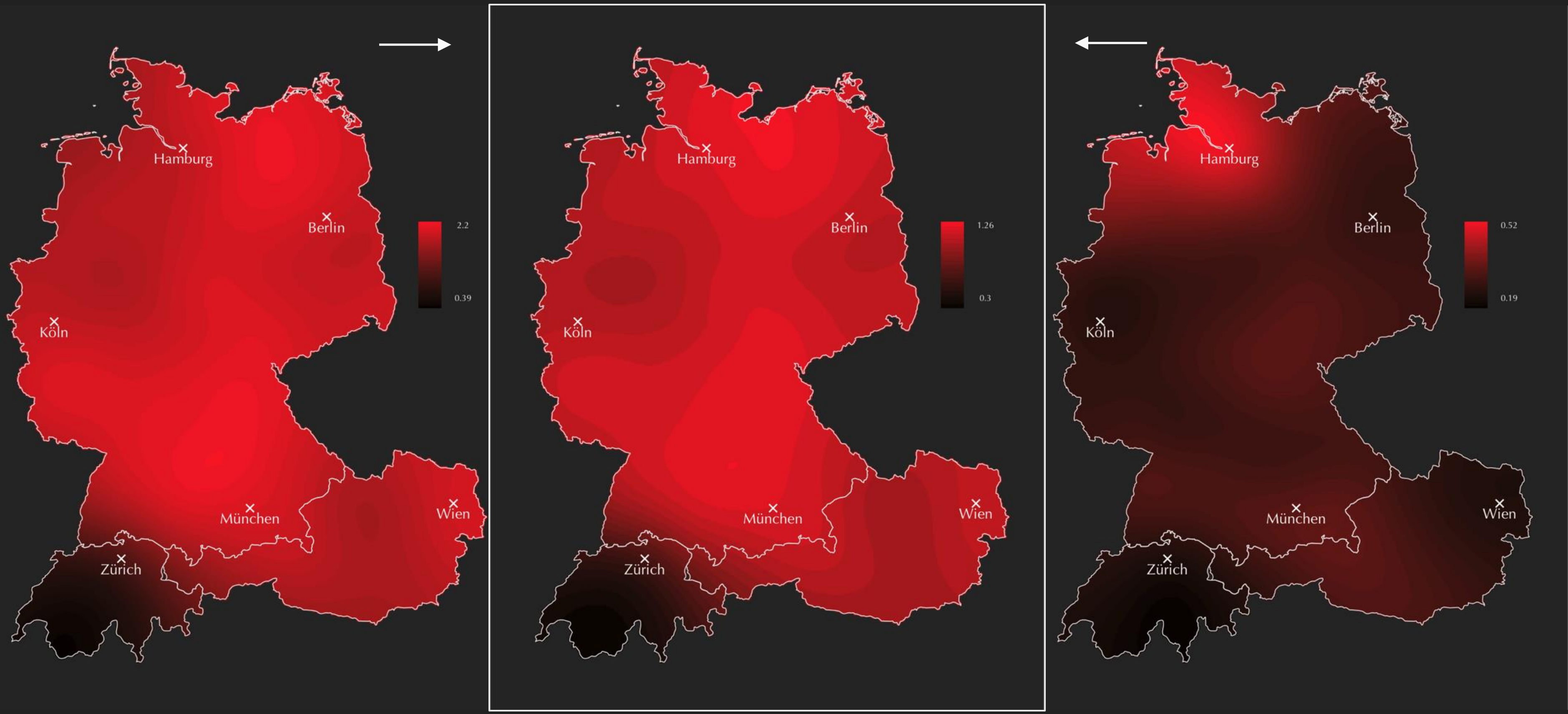
- The maps are a visualisation of areas of high / low feature use (in this corpus)

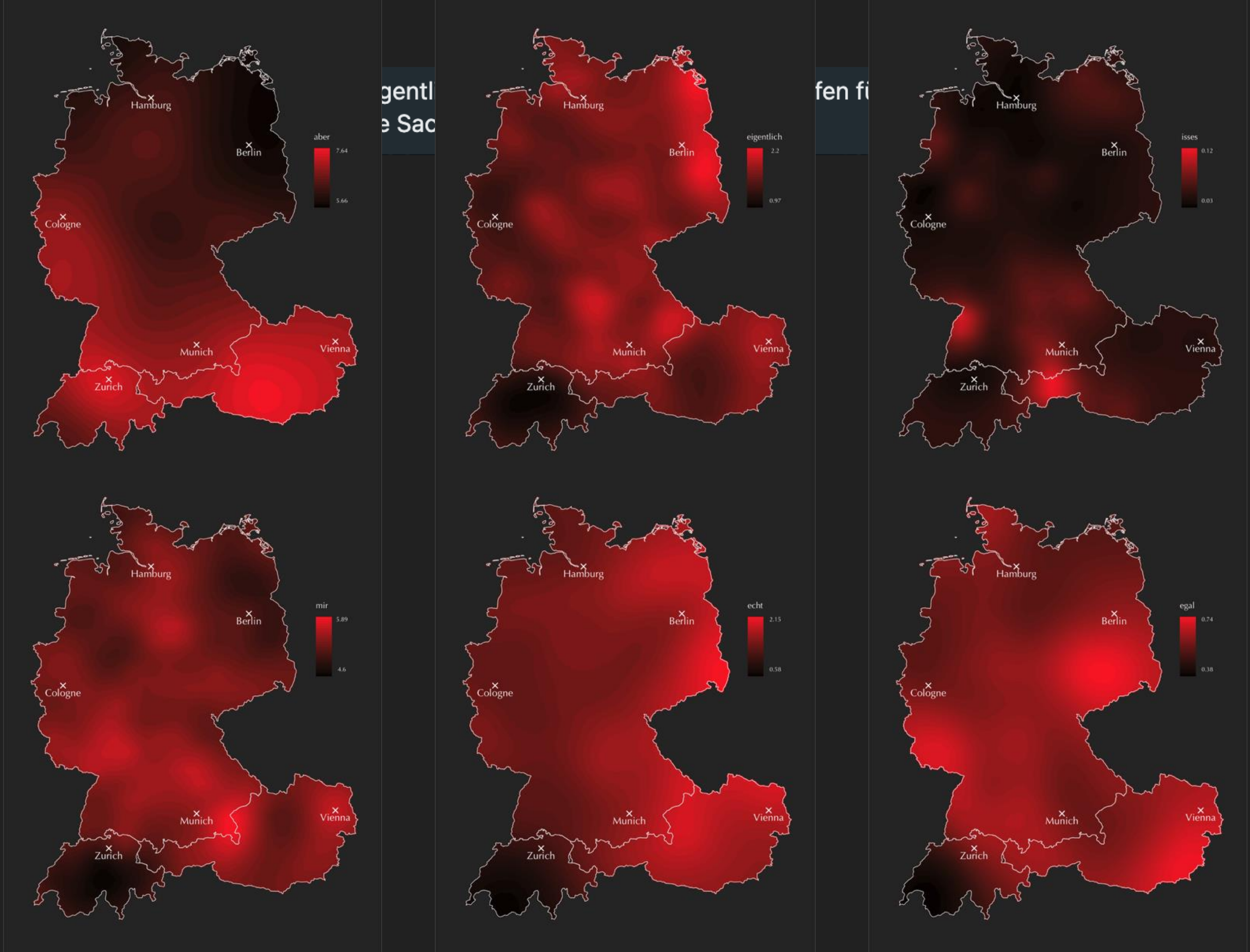
→ Prediction of country

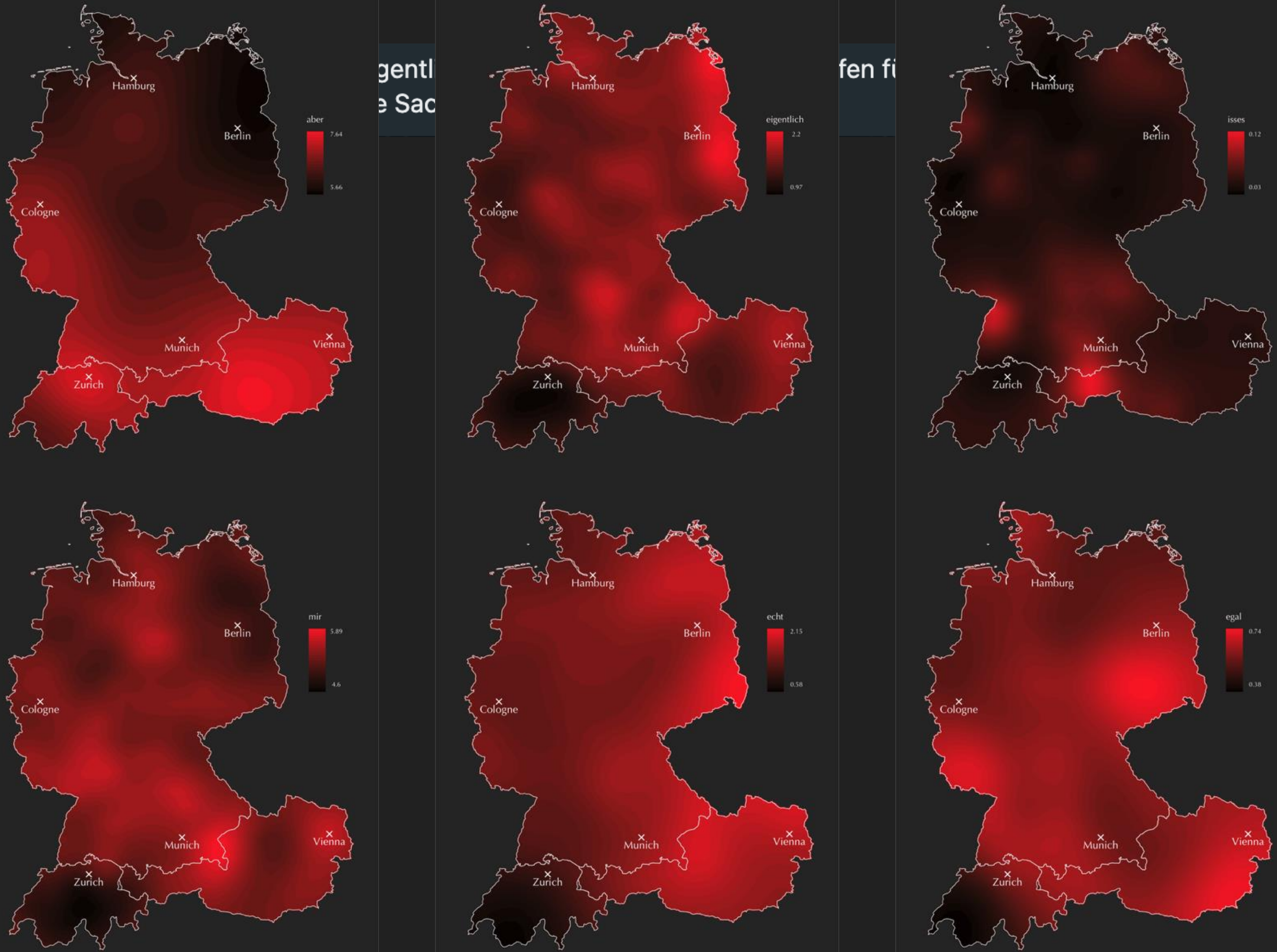
→ Prediction of location (more narrowly)



“keine polizei“ (*no police*)




Hamburg
Berlin
Cologne
Munich
Vienna
Zurich
aber
7.64
5.66
gentl
e Sac
eigentlich
2.2
0.97
fen fü
isses
0.12
0.03
mir
5.89
4.6
echt
2.15
0.58
egal
0.74
0.38



Hamburg
Berlin
Cologne
Munich
Vienna
Zurich
aber
7.64
5.66
eigentlich
2.2
0.97
isses
0.12
0.03
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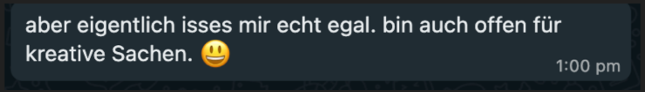
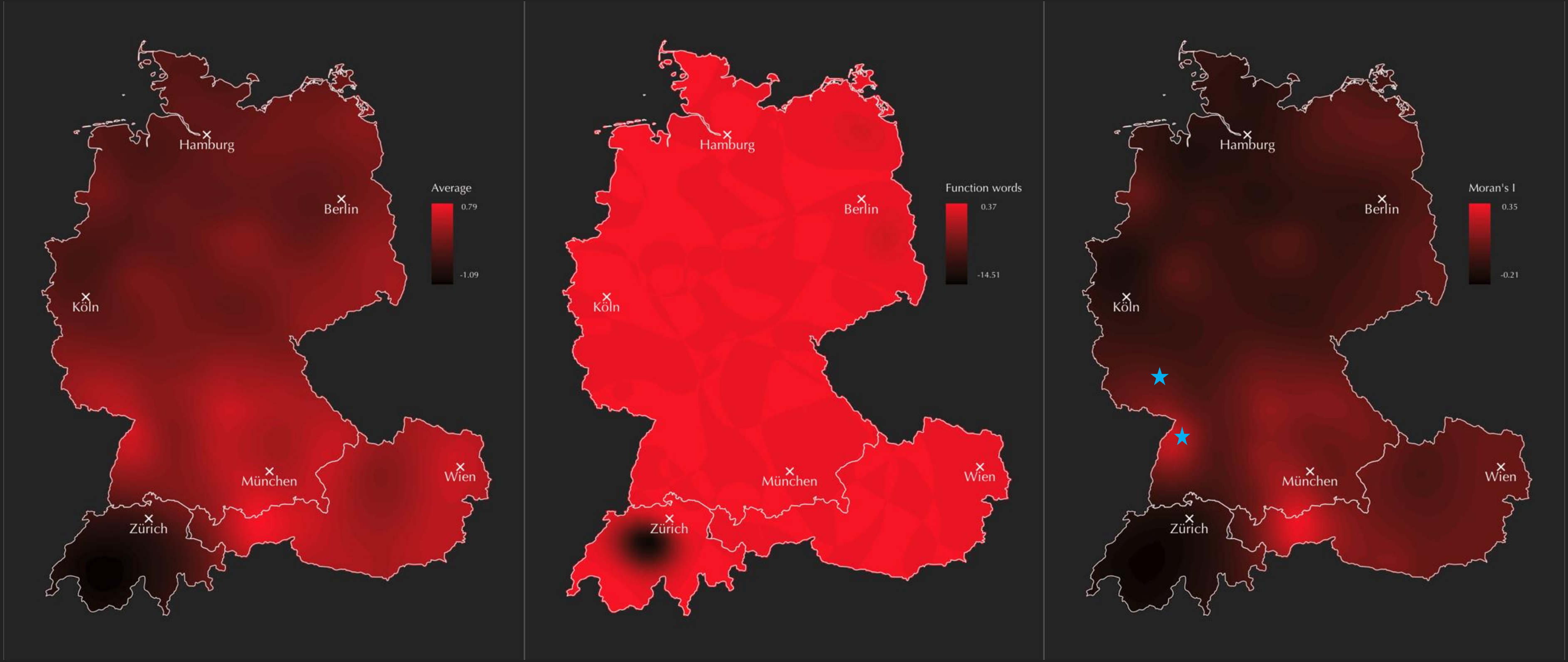
echt
2.15
0.58
egal
0.74
0.38


aber eigentlich isses mir echt egal. bin auch offen für kreative Sachen. 😃
1:00 pm


Hamburg
Berlin
Köln
München
Wien
Zürich
Average
0.79
-1.09
Function words
0.37
-14.51
Moran's I
0.35
-0.21

aber eigentlich isses mir echt egal. bin auch offen für kreative Sachen. 😃 1:00 pm

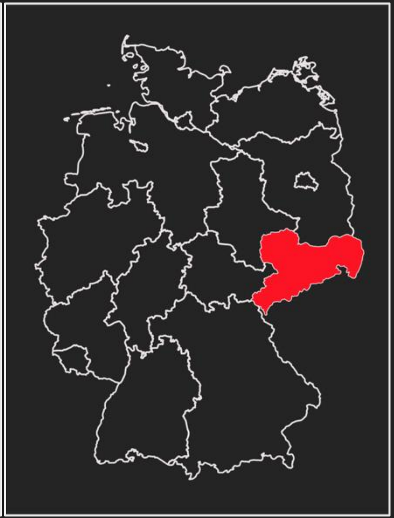
- First step: prediction of country DE
- Second step: more narrow prediction
  - for place of socialisation (20 years)
  - for place of living (15+ years)

kennt ihr diesen Hans Entertainment Moment, wenn du Niveau in einem anonymen messageboard erwartest? ich nicht.

Do you know that Hans Entertainment moment when you expect class on an anonymous message board? I don't.

Country: DE

Within country:

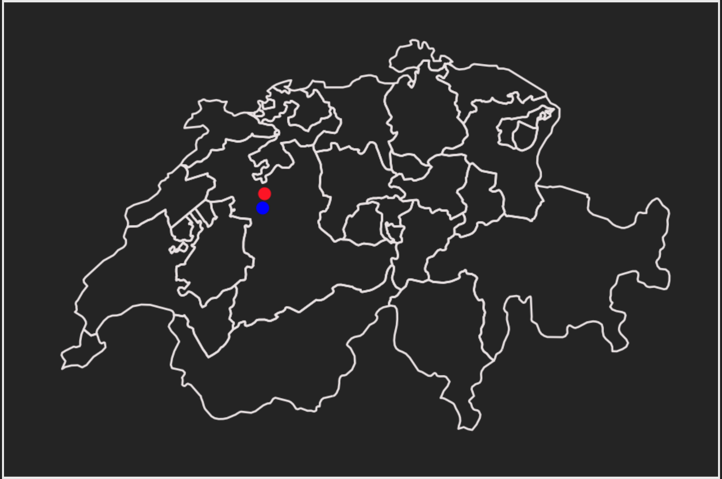


iviu Fieber hesch? Hesch gmässe? Wed Fieber hesch würdi itz auso no chly warte. Sobauds Fieber wäg isch chasch ja ga

Do you have a fever? Did you take [your temperature]? If you have a fever I would wait a bit. As soon as the fever is gone you can go.

Country: CH

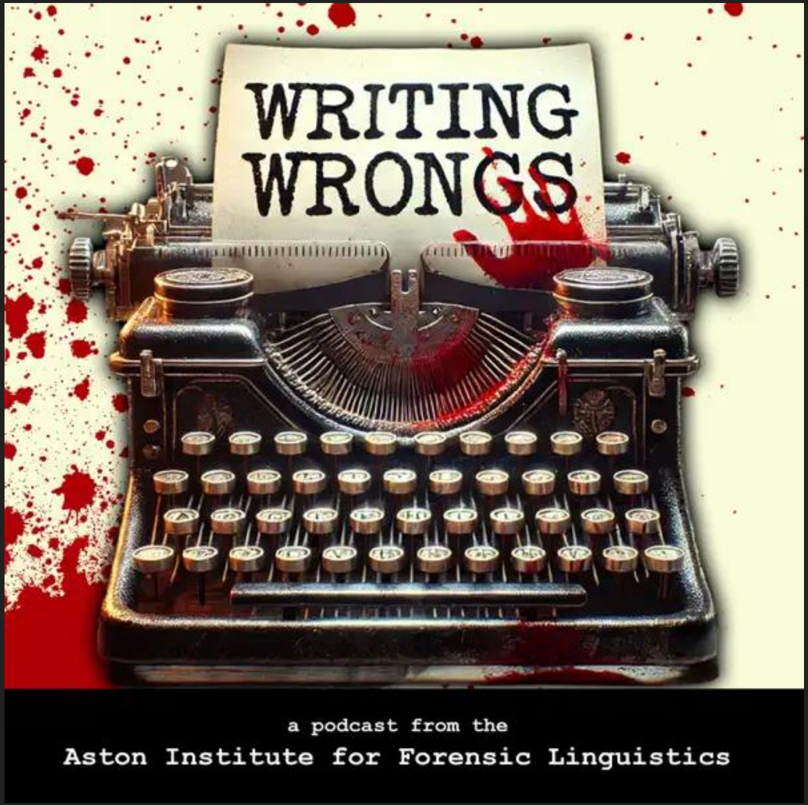
Within country:



Mapping

Roemling, D., Winter, B., & Grieve, J. (2025). Visualizing map data for linguistics using `ggplot2`: A tutorial with examples from dialectology and typology. *Journal of Linguistic Geography*, 1-15. doi:10.1017/jlg.2024.11

Writing Wrongs



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Images

- https://thelegalseagull.com/cdn/shop/products/5_thumbnail_1200x1200.jpg?v=1568344894
- https://www.streetoncriminallawyers.com.au/wp-content/uploads/2022/11/43-2.png
- https://cdn-icons-png.flaticon.com/512/293/293906.png
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- https://res.cloudinary.com/aenetworks/image/upload/c_fill,w_1200,h_630,g_auto/dpr_auto/f_auto/q_auto:eco/v1/discovery-shows-early-christians-didnt-always-take-the-bible-literallys-featured-photo?_a=BAVAZGDX0
- https://res.klook.com/images/fl_lossy.progressive,q_65/c_fill,w_1200,h_630/w_80,x_15,y_15,g_south_west,l_Klook_water_br_trans_yhcmh3/activities/ffnm1b0qm18goa0igxi0/Jack%20The%20Ripper%20Tour%20%3A%20Solve%20The%20Crime.jpg